

Q. Does trade liberalization benefits everyone? ←

Who are the winners / losers?

— Mostly a repeat of previous class —

503 Development

L-12

9/04/2025

Climate Change, Vulnerability
& Basic Understanding

* Paris Agreement, Green job, green finance,

✓ LDC graduation → The criteria of graduation and deferring

✓ The Relationship between disasters and sectoral and National GDP.

→ Source: Benson and Clay (2004)

* Extreme weather event.

(*) Why is Drought less talked about → ^{drought is more} manageable

⇒ Disaster vs. Hazard
↓
Management Failure

* Cropping intensity

* Time frame & water level
May 2nd week for Boro
(Haor belt)

	1940s	2020s	Seeding	Harvest
Aman	80%	98%	June-July	Nov / Dec
Aus	17%	2%	Feb / Mar	Sep / Oct
Boro	3%	60%	Dec	May (2nd week)

change started during the 80's

Reason: Development of irrigation system

1974 → The famine caused large scale changes in the economy and geo-politics of BD.

<PL 420> The US food aid programme. Giving the excess supply of food to LDCs in exchange of favorable (US) political contracts.

→ 1975 - Remittance started flowing

→ 1976-79 → Drought

* Oil export from Jordan during the 70's → Read up.

BARD → Research of ~~agri~~ agriculture. Was established for "rice-research"

BADC → To circulate irrigation tech and new seed

* Cyclone and water surge.

□ Benson & Clay the pattern (why flat after 1980)

* Case Study of 1998 Flood.

1. In FY 1999, Revenue fell short of target by 5.2%.
2. Expenditure exceeded target by 5.2%.
3. Exacerbating budget deficit on current account
4. For the first time in recent years expenditure under ADP exceeded its target by 2.4%.
5. This surge in ADP exp. in part reflected the demand of post-flood rehab. Program, where expenditure for repairing damage to infrastructure was needed to be stepped up
6. Overall deficit in public exp. rose by 52.2 Bn Tk.
7. Deficit was financed through a combination of enhanced deficit financing through the banking system & foreign Aid.

* Invest for Climate Change. Why the Delta should be concerned for climate change?

conduct

IPCC Assessment warns —

With current level of CO_2 , the world face a net change of 0.1°C per decade over the next 50 years.

→ Nobel Prize in 2007.

→ After that temp will increase at 0.5°C per decade rate.

→ BD agriculture → One of the highest greenhouse gas emitter in the world (within agriculture)

মিথেন warming potential is 24, which is a byproduct of Boro & Aman rice production

→ How does a cyclone form?

তিন মণ্ড অঞ্চল সংকেত : অল্পতর মাঝে স্থানীয় অধিকাংশ
(on the surface)

দশম " " " : মহাবিকল সংকেত
Super cyclone

জোয়ার জোলে → মাঝে মাঝে জোয়ারের প্রথম সত্তাহ

↓ মোক নোহোঁর পর্যন্ত

জলদস্যু (সাহা) fishing trip lasts upto 15 days.

→ no institutional ban, only 'মহাজল' with upto 200% interest-rate.

① ৬ মণ্ড সংকেত পোনে অবশ্যই উন্নয়নে ফিরে আসতে হবে,

→ Climate change → Higher rate of danger signals → fishermen has to come back with have done fishing trip.
Lower fishing harvest.

→ During the last 50 years Bay of Bengal faced a temp rise of 0.6°C (compared to pre industrial)

→ তিন থেকে ছয় দশক যিনি সংকটের সম্মুখীন
আন্তর্জাতিক স্থানীয় সংস্থা ইতি। (জুন - অক্টোবর)

Avoid the Unmanageable

Manage the Unavoidable

→ Mitigation

→ Adaptation

● Prevention

Preparation for the worst

↓
Reducing emission

↓
Adjusting cropping

↓
Managing GHGs

↓
for flood

□ Mechanism of River Flood

⇒ Ganges, Brahmaputra, Meghna → The Big Three
(Padma) (Jamuna)

54 rivers → common with India

↳ land mass 7% in BD.

↳ 93% water flows through BD

⇒ What is harm and risk

$$\text{Vulnerability} \propto \int (E * S * R^{-1})$$

Hazard exposure ↓ Sensitivity of System Resilience of the System

Risk: A condition that puts a system into vulnerability.

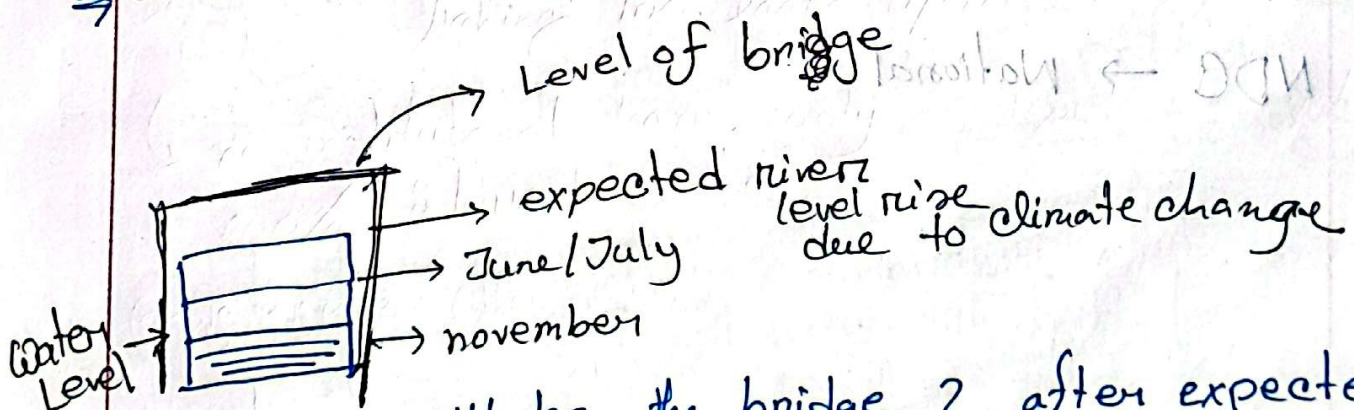
Harm: Net adverse effect, in this case of climate stimuli.

Our object: To respond toward navigating and preventing harm.

⇒ Climate Parameters

Flood, drought, Salinity, Cyclone storm surge
Other Parameter.
Erosion, land-slide etc.

⇒ Govt. Investment on infrastructure & Disaster/Climate change.



⑨ Where will be the bridge? after expected climate change on regular flood level? Cost-Benefit Analysis.

Financing Disaster

- Additionality
- Full-cost

□ Fundamentals of Paris Agreement

UNFCCC

- ① Principle of equity and common but differentiated responsibilities and respective capability.
- ② Recognizing the need of an effective and progressive response to the urgent threat of climate change on the basis of the best available scientific knowledge.
- ③ Also recognizing the specific needs and special circumstances of developing country parties.

NDC → National Determined Contribution



Paris Agreement (2015)

Kyoto Protocol → Mitigation not Adaptation. (1992)

CBDR with RC

↓
Common Band Differential Responsibility with Respective Capability

1990 → Temp rise 1.5°C control
base year.

Pre-industrial Era T_1 → $+1.5^{\circ}\text{C}$ control was demanded by LDCs

1990 T_2 → $+1.5^{\circ}\text{C}$ agreed by industrial nations.

→ To do that GHG must be controlled.

INDC mechanism → to control GHG.

↓
intended nationally determined commitment

From PA (2015) this became: Nationally Determined Contribution

G77 + China → India, Brazil, Mexico, S. Africa.

Rio-Summit '92 → Rio charter → Kyoto protocol

Only historical testimony of Developed countries that they are responsible for GHG emission & climate change
* agreed to compensating affected nations.

Paris Agreement (2015) → This compensation was Retracted.

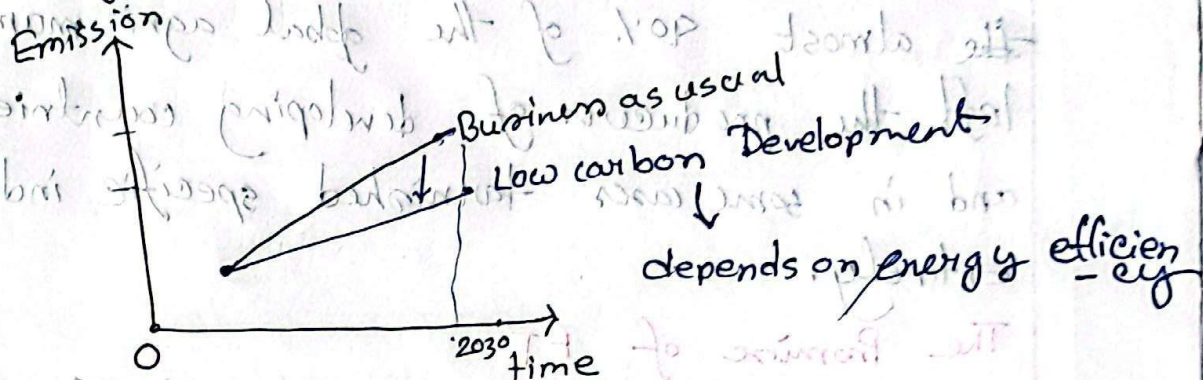
Ref → PARIS Agreement Article-1 उपरोक्त वाक्यांश
लेखक लाई : 810

Climate Change — [Anthropogenic → natural
Accelerated → since IR

2007 → first recognition of human effect of Climate change. Talk about adaptation with Mitigation started.

Dr. Anil Agarwal (Ind) } not sum GHG but per
Dr. Atik Rahman (BD) } capita emission that matters.

→ 2005: How to reduce per capita GHG
Right to Develop and GHG emission



(PEA) 2015

④ Emphasizing

⑤ Acknowledging

→ Differential Vulnerability
↓
Disaster management.

Inter generational
Equity

→ Leaving resources for future generation as such they can maintain the same level of life-style as the present generation.